

VAISHALI SONKER

AI/ML | Generative AI | Software Development | Data Analytics
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Professional Summary

Computer Science undergraduate with hands-on experience in Generative AI, Machine Learning, and backend development. Skilled in Python, SQL, FastAPI, RAG, LLMs, FAISS, PostgreSQL, and LangGraph, with experience building AI-powered applications and data-driven solutions. Strong foundation in problem-solving, software development, and machine learning.

Technical Skills

Languages: Python, SQL, Java, JavaScript, TypeScript

AI / ML: Scikit-learn, XGBoost, Regression, Classification, Time Series, SARIMA, Feature Engineering, Hyperparameter Tuning

Generative AI: RAG, LLMs, Gemini, Groq, LangChain, LangGraph, LLM Integration, Prompt Engineering, Embeddings, Semantic Search, Context-Aware Memory, LLM Orchestration, Multimodal AI

Backend / APIs: FastAPI, REST APIs, Uvicorn, Pydantic, SQLAlchemy, Async Programming, WebSockets, API Integration

Data / Databases: Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Power Query, ETL, EDA, Data Cleaning

Computer Vision / Multimedia: OpenCV, OCR, Tesseract, FFmpeg, Speech-to-Text, Whisper

Tools: React, Next.js, Streamlit, Power BI, Excel, Git, GitHub, Cloudinary, Alembic

Experience

Machine Learning Intern | **CNE Associates**

Dec 2025 – Jan 2026

- Developed classification and regression models using Python and Scikit-learn, improving accuracy by **10–15%** through feature engineering and hyperparameter tuning.
- Built ML pipelines across **3+ datasets** for preprocessing, training, and evaluation, reducing data inconsistencies by **~20%**.
- Evaluated models using **F1-score and RMSE** and documented reproducible experiments for iterative model improvement.

Key Projects

VidyaRoom — AI-Powered Lecture Assistant **GitHub** | *Python, FastAPI, Next.js, PostgreSQL, Groq, Gemini, WebSockets*

- Built an AI lecture platform with real-time transcription, translation, topic detection, AI Q&A, and analytics using Whisper, Gemini, and Groq.
- Developed asynchronous FastAPI APIs with PostgreSQL, SQLAlchemy, WebSockets, authentication, FFmpeg, Cloudinary, and OCR-based processing.

Vera — AI Resume Assistant **GitHub** | *Python, FastAPI, React, Gemini, FAISS, Google Sheets*

- Developed a RAG-based AI assistant using Gemini embeddings, FAISS vector search, semantic retrieval, prompt engineering, and context-grounded LLM responses.
- Integrated FastAPI with React for streaming responses, Google Sheets feedback collection, and SMTP-based contact functionality.

Chat with PDFs — RAG-Based Document Q&A | *Python, FastAPI, FAISS, SentenceTransformers, Gemini*

- Built an end-to-end RAG pipeline using document chunking, SentenceTransformer embeddings, FAISS similarity search, and Gemini.
- Reduced LLM token usage by **~98%** by retrieving only the top-5 relevant document chunks per query.

Sales Forecasting Dashboard **GitHub** | *Python, SARIMA, XGBoost, Streamlit, Pandas*

- Developed SARIMA and XGBoost forecasting models, reducing RMSE by **18%** through comparative model evaluation.
- Engineered lag, rolling, and date-based features and deployed a Streamlit dashboard for **30–60 day** forecasts.

Sales KPI Dashboard **GitHub** | *Power BI, Excel, Power Query*

- Built a Power BI dashboard for revenue, profit, regional performance, and business KPIs using dynamic filters and drill-downs.
- Transformed data from **5+ regional sources** using Power Query, reducing manual reporting effort by **~40%**.

Achievements

- Best Approach Award — IBM BOB Hackathon** for the proposed technical approach and solution.

Education

Bachelor of Technology (B.Tech) — Computer Science Engineering | **CPI: 7.55/10**

Expected: 2027

4th Year Undergraduate